

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 j_grid = 1001      # number of grid points
5 n_grid = 3001      # number of time steps n
6
7 C = 3/2            # CFL condition = c*Δt/Δy
8
9 source_1 = 400      # source location of EM wave Case 1
10 source_2 = 10       # source location of EM wave Case 2
11 source = source_2
12
13 Ez = np.zeros(j_grid)      # Electric field (E_z)
14 Hx = np.zeros(j_grid - 1)  # Magnetic field (H_x)
15
16 Ez_prev_left = 0.0         # store previous value of Ez on the left
17 Ez_prev_right = 0.0        # store previous value of Ez on the right
18
19 for n in range(n_grid):    # loop over time
20     for j in range(j_grid - 1):    # loop over space in defined grid
21         Hx[j] += (Ez[j+1] - Ez[j]) * C    # update Hx based on previous Ez
22
23     for j in range(1, j_grid - 1):
24         Ez[j] += (Hx[j] - Hx[j-1]) * C    # update Ez based on previous Hx
25
26     # Boundary condition for absorbing (Mur boundary condition)
27     Ez[0] = Ez_prev_left + (C - 1) / (C + 1) * (Ez[1] - Ez[0])    # define for left boundary
28     Ez_prev_left = Ez[1]    # store for next iteration
29     Ez[-1] = Ez_prev_right + (C - 1) / (C + 1) * (Ez[-2] - Ez[-1])    # define for right boundary
30     Ez_prev_right = Ez[-2]
31
32     source_function_1 = np.sin(2 * np.pi * n * C / 20)    # source function Case 1
33     source_function_2 = np.exp(- ((n / 30) - 5) ** 2)    # source function Case 2
34
35     Ez[source] = source_function_2
36
37     if n == 500 or n == 1000 or n == 3000:    # PLOT at timesteps 500, 1000 and 3000
38         plt.figure()
39         plt.plot(Ez)
40         plt.title(f'Timestep {n} CFL = {C}')
41         plt.xlabel('Grid point')
42         plt.ylabel('Ez')
43         if Ez[source] == source_function_1:
44             plt.suptitle('Case 1')
45             plt.savefig(f'case1_CFL{C}_time{n}.png')
46         elif Ez[source] == source_function_2:
47             plt.suptitle('Case 2')
48             plt.savefig(f'case2_CFL{C}_time{n}.png')
49 plt.show()

```